Foundations of scientific reading

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Attendance:

https://checkin.bangor.ac.uk/

Agenda

- The structure of an empirical article
- Background to the study: two strategies for taking a penalty kick
- Reading the article: Introduction and Method
- Reading the article: Results, and how statistical results are reported
- Reading the article: Discussion, and implications for practice

Section	Question it answers

The structure of an empirical article

Section	Question it answers
Title and authors	What is the study about, and who conducted it?
Abstract	What was done and found, in about 200 words?
Introduction	What is already known, what is not, and what does this study ask (aims and hypotheses)?
Method	Who took part, what was manipulated and measured, and how?
Results	What was found, usually with statistical tests, tables and figures?
Discussion	What do the findings mean, how do they relate to theory, and what are the limitations and implications?
References	Which earlier work does the study build on?

An efficient order of reading

1. Title and abstract: the overall picture.
2. Final paragraph of the Introduction: the aim and the predictions.
3. Method: participants, design, independent and dependent variables.
4. Tables and figures in the Results, and the sentences that describe them.
5. First and final paragraphs of the Discussion: the main conclusion and its implications.
6. The remaining text, for detail.

Structured reading sheet

Structured reading sheet. Used in the reading tasks today and useful for revision.

Background to the paper

Two strategies when taking a football (soccer) penalty, depending on whether the movements of the goalkeeper are considered

Keeper-independent: ignore the goalkeeper

- Before the run-up, choose the target location
- Aim at the target location
- Shoot at the target location regardless of the goalkeeper's actions

Keeper-dependent: attend to the goalkeeper

- Do not choose a target location before the run-up
- During the run-up, anticipate the direction of the goalkeeper's dive
- Shoot in the opposite direction

Discussion in small groups: **which strategy is preferable, and why?**

Arguments in favour of the keeper-independent strategy

Navarro and colleagues summarise three arguments:

1. **There is insufficient time for a keeper-dependent strategy**

If the goalkeeper moves within approximately 400 ms of foot–ball contact, kickers are less likely to place the ball in the empty half of the goal, and their kicks are less accurate.

2. **The keeper-independent strategy allows visual focus on the target**

It permits more adaptive gaze behaviour: fixating the target provides visual information, and information about eye position, that supports accurate aiming (see the quiet eye, in the session on perception).

3. **Shots to the upper corners are very unlikely to be saved**

Descriptive analyses of international competitions indicate that goalkeepers did not save shots directed at the upper corners.

Biomechanical studies indicate that a kick to an upper corner at a sufficient speed (above approximately 22 m/s) cannot be intercepted.

Can the goalkeeper really be ignored?

If these arguments are accepted, the keeper-independent strategy should be adopted. But can it be adopted fully? **Can a penalty kicker ignore the goalkeeper?**

The study by Navarro et al. (2013) addresses this question.

While reading the Method, compare the designs choices with what you would have done in your own study: which variables did the authors control? Which did they not? Which variables would you have controlled, and how? What are the advantages and disadvantages of the authors' choices?

Assessment 1

The paper by Navarro et al. (2013) is the basis of assessment 1.

A one-hour online test of 20 multiple-choice questions, on Blackboard. It opens after today's session. The deadline and full details are in the module handbook.

Navarro, M., van der Kamp, J., Ranvaud, R., & Savelsbergh, G. J. P. (2013). The mere presence of a goalkeeper affects the accuracy of penalty kicks. *Journal of Sports Sciences, 31*(9), 921–929. https://doi.org/10.1080/02640414.2012.762602

Demonstrations:

1. Where to find this information in the module handbook, including the deadline

Can the deadline be extended? Yes, where there are valid reasons. Contact your personal tutor.

2. How to access the full paper through the Talis reading list
3. How to complete assessment 1

To obtain the highest marks, the slides are not sufficient: they are not exhaustive. **The paper itself should be studied**.

Read the paper: Introduction and Method

10 minutes. Complete sections 1–4 of the reading sheet.

Focus on:

- What were the aims of this study?
- Who were the participants, and how were they selected?
- What was manipulated (independent variable)?
- What was measured (dependent variables)?

Review questions: Introduction and Method

Where in an empirical article is the aim of the study normally stated?

a) In the references
b) At the end of the Introduction
c) In the first table of the Results
d) Only in the title

Correct response: b

In a repeated-measures (within-participant) design:

a) each participant is tested in only one condition

b) each participant is tested in every condition
c) participants choose their condition
d) there is no independent variable

Correct response: b

Why might the order of conditions be counterbalanced across participants?

a) To increase the number of participants
b) To prevent effects of practice or fatigue from being confounded with the conditions
c) To make the conditions more difficult
d) To avoid the need for a statistical test

Correct response: b

Only participants who passed a skill test took part. What is one consequence?

a) The results cannot be analysed
b) The findings apply most directly to skilled players and may not generalise to other populations
c) The design becomes between-participants
d) The dependent variable changes

Correct response: b

The experimental set-up

Each upper corner of the goal contained a target area, with a target circle at its centre.

The three conditions used in the study

Instruction: hit the target area indicated by the experimenter before each kick (left or right upper corner), with sufficient power, using a keeper-independent strategy.

Conditions (every participant performed all three):

A) no goalkeeper

B) goalkeeper

C) knowledgeable goalkeeper: the kicker told the goalkeeper the intended target before the kick

Performance coding used in the study

hit: the ball enters the intended target area

miss: the ball enters the goal, outside the target area

failure: the ball does not enter the goal

saved: the ball is blocked by the goalkeeper

not saved: the ball is not blocked by the goalkeeper

Dependent variable: the number of kicks in each category

Other dependent variables

Absolute error: the distance between where the ball crossed the goal line and the centre of the target (or the centre of the goal); an index of **accuracy**

Variable error: the variability of ball locations across repeated kicks; an index of **inconsistency**

Ball speed: the distance travelled divided by the flight time

Read the paper: Results

10 minutes. Complete section 5 of the reading sheet.

Focus on:

- How did the numbers of hits, misses and failures differ across the three conditions?

- How did the number of saves differ between the two goalkeeper conditions?
- How did absolute and variable error differ across the three conditions?

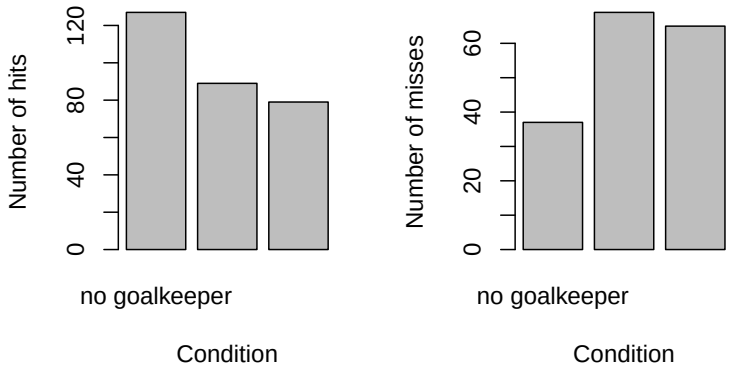
How statistical results are reported

The study used statistical tests other than the t-test, but the logic is identical: a statistic value is compared with the values expected by chance if there were no true difference.

Reported result	Reads as
$^{2}(4, N = 600) = 26.64, p < 0.001$	chi-square test on counts of 600 kicks: the distribution of outcomes differed significantly across conditions
$F(2,18) = 7.87, p < 0.01$	analysis of variance across three conditions: absolute error differed significantly across conditions
$p = 0.054$, "marginally significant"	just above the conventional .05 criterion

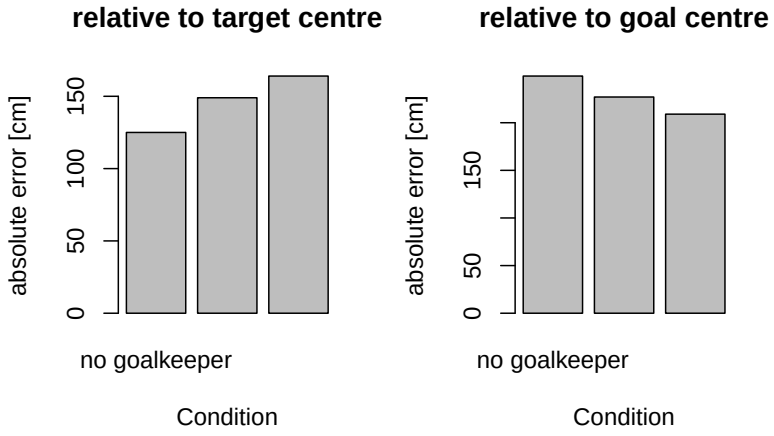
In Table I, superscript letters mark which conditions differed significantly from one another.

Some results: number of hits and misses



In the no-goalkeeper condition, the number of hits was significantly higher, and the number of misses significantly lower, than in the goalkeeper and knowledgeable-goalkeeper conditions.

Some results: absolute error



Relative to the target centre, absolute error was larger in the knowledgeable-goalkeeper condition than in the other two con-

ditions. Relative to the goal centre, it was smallest in the knowledgeable-goalkeeper condition: shots were the most centralised, that is, closest to the goalkeeper.

Variable error did not differ across conditions: the shots were consistently displaced towards the goalkeeper, not merely more scattered. In the terms of the session on data and measurement: a systematic error, not a random one.

Review questions: Results

A result is reported as "$p < 0.05$". This indicates that:

a) the effect is large and important for performance
b) a difference this large would rarely arise by chance if there were no true difference
c) fewer than 5% of participants showed the effect
d) the measurement was reliable

Correct response: b

A chi-square test is appropriate for comparing conditions when the dependent variable is:

a) a mean distance in centimetres
b) the number of observations falling into each category
c) a reaction time
d) a correlation between two measures

Correct response: b

In a results table, two conditions share the same superscript letter. This indicates that:

a) the two conditions differed significantly
b) the two conditions did not differ significantly from each other
c) the values were estimated
d) the conditions were tested on the same day

Correct response: b

A study reports $p = 0.054$ and describes the effect as "marginally significant". Using the conventional criterion, the effect is:

a) statistically significant
b) not statistically significant
c) large
d) caused by an error in measurement

Correct response: b

Reading critically

- **Ball speed.** Participants were instructed to exceed approximately 22 m/s, yet mean speeds were 17.5–18.8 m/s in all conditions (Table I). Only kicks below 22 m/s were saved.
- **Participants.** Ten participants were selected from 27 by a pretest of aiming accuracy. The findings concern skilled amateur players.
- **Setting.** The kicks were taken in an experiment, without the pressure of competition.

How might each point affect the conclusions that can be drawn?

Read the paper: Discussion

10 minutes. Complete sections 6–8 of the reading sheet.

Focus on:

- How do the authors explain their results? Which theory do they use?
- Which alternative explanation do they consider?
- Which applied recommendations do they make, for example for practice or coaching?

The authors' explanations

- **Response activation model** (primary account). Before an aiming action, both the target and salient non-targets in the environment activate competing responses. A salient non-target that cannot be fully inhibited biases the movement towards it. The goalkeeper acts as such a non-target.
- **Ironic processes** (alternative account). An instruction to ignore the goalkeeper may, ironically, increase attention to the goalkeeper.

Links with other topics: salient stimuli capture attention (the session on attention); the preparation and control of aiming actions (the session on action and motor control).

Review question: Discussion

In which section of an empirical article are the limitations of a study and the directions for future research normally reported?

a) Abstract
b) Method
c) Results
d) Discussion

Correct response: d

Practical take-aways (based on this paper)

1. How would you advise a coach who asked for a recommended penalty strategy?

The study supports the keeper-independent strategy as the more secure choice, but shows that it is difficult to adopt fully. The authors recommend practising in the presence of a goalkeeper, and training gaze control (e.g., quiet eye training), so that

gaze is directed at the target and the ball rather than at the goalkeeper.

2. Example. 5 July 2014, quarter-final of the 2014 FIFA World Cup in Brazil: Netherlands – Costa Rica.

The Guardian video about Krul's experience at the World Cup